

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	259019	51.791370
1	[Git, Change, Create]	82087	16.413461
2	[Git, Change, Delete]	50533	10.104175
3	[Git, Change, Rename]	24069	4.812645
4	[File, Git]	16055	3.210230
5	[Git, Commit]	15712	3.141646
6	[Java, ByteCode, Parameter]	9258	1.851156
7	[Java, ByteCode, Member, Method]	8914	1.782372
8	[Java, ByteCode, Bound]	6308	1.261297
9	[Java, Value, ByteCode, Annotation]	5351	1.069943
10	[Java, ByteCode, ParameterizedType, Bound]	4854	0.970567
11	[Xml, Element]	2404	0.480685
12	[Java, ByteCode, Member, Field]	2197	0.439295
13	[Xml, Text]	1612	0.322323
14	[Java, ByteCode, Member, Constructor, Method]	1577	0.315324
15	[Java, ByteCode, Bound, WildcardType]	969	0.193753
16	[Java, ByteCode, Member, Method, Lambda]	851	0.170159
17	[Java, ByteCode, Bound, TypeVariable]	831	0.166160
18	[Type, File, Java, ByteCode, JavaType]	689	0.137767
19	[Type, File, Java, Class, ByteCode]	651	0.130169
20	[Json, Key]	544	0.108774
21	[Value, Json, Scalar]	534	0.106774
22	[Git, Change, Copy]	478	0.095577
23	[Type, File, Java, ByteCode, ResolvedDuplicate...]	467	0.093378
24	[Java, ByteCode, Member, Method, GenericDeclar...]	456	0.091178
25	[Java, Value, ByteCode, Primitive]	326	0.065184
26	[Author, Git, Person]	303	0.060585
27	[Type, File, Java, ByteCode, ExternalType]	302	0.060386
28	[Committer, Git, Person]	246	0.049188
29	[Maven, Dependency]	194	0.038791

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

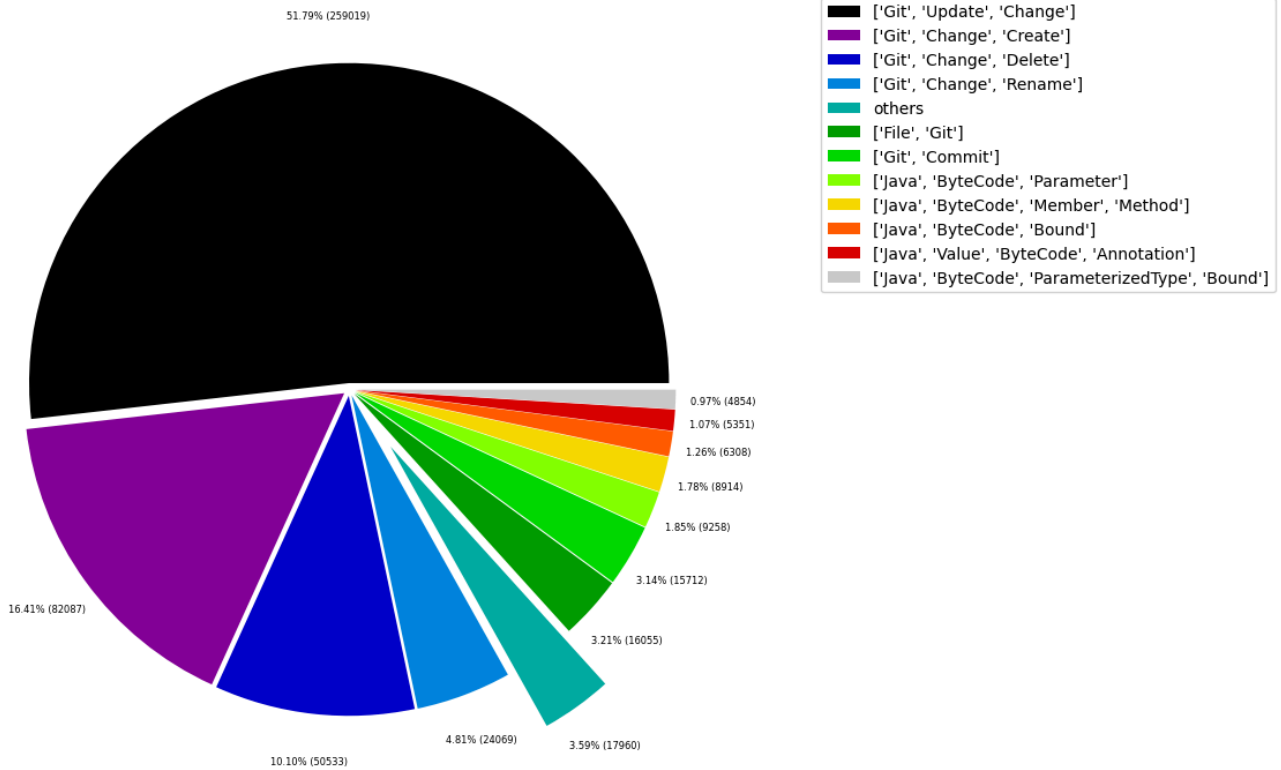


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000200
1	[Git, Branch]	1	0.000200
2	[Repository, File, Git]	1	0.000200
3	[File, Json]	2	0.000400
4	[File]	3	0.000600
5	[Type, File, Java, ByteCode, GenericDeclaratio...	4	0.000800
6	[Maven, Exclusion]	4	0.000800
7	[Type, File, Java, ByteCode, Throwable, Extern...	8	0.001600
8	[Value, Json, Array]	8	0.001600
9	[Java, ByteCode, Member, Constructor, Method, ...	9	0.001800
10	[Maven, Scm]	11	0.002199
11	[File, Maven, Xml, Pom, Document]	11	0.002199
12	[Type, File, Java, ByteCode, Throwable, Resolv...	11	0.002199
13	[Type, File, Java, ByteCode, Void]	11	0.002199
14	[Java, ManifestSection]	11	0.002199
15	[File, Java, Manifest]	11	0.002199
16	[File, Artifact, Jar, Archive, Zip, Java]	11	0.002199
17	[File, Java, ServiceLoader]	12	0.002399
18	[File, Java, Properties]	13	0.002599
19	[Maven, ExecutionGoal]	13	0.002599
20	[Maven, PluginExecution]	13	0.002599
21	[Type, File, Java, ByteCode, Enum]	17	0.003399
22	[jQAssistant, Rule, Concept]	19	0.003799
23	[Maven, Configuration]	20	0.003999
24	[Maven, Plugin]	20	0.003999
25	[Xml, Attribute]	22	0.004399
26	[Type, File, Java, ByteCode, PrimitiveType]	39	0.007798
27	[Type, File, Java, ByteCode, Annotation]	42	0.008398
28	[Type, File, Java, Class, ByteCode, Throwable]	42	0.008398
29	[Xml, Namespace]	44	0.008798

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

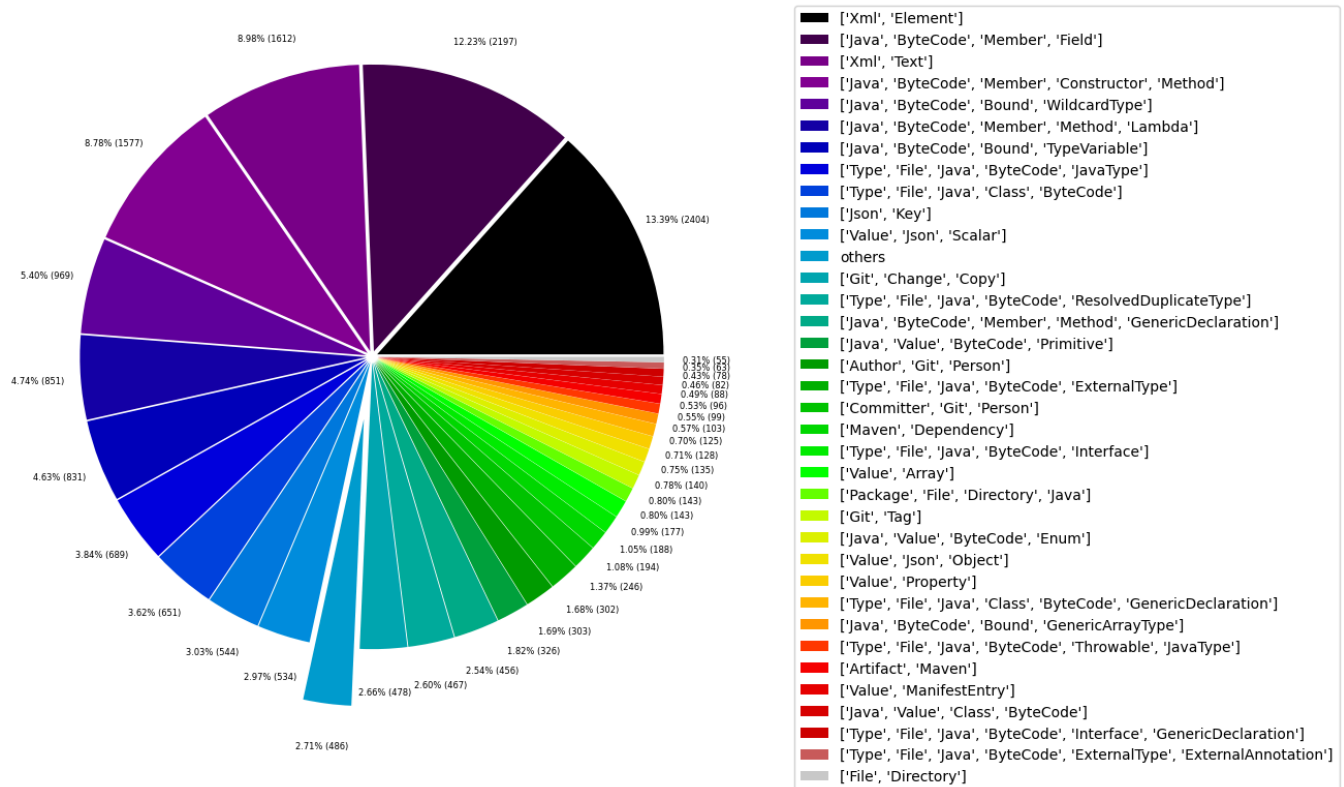


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

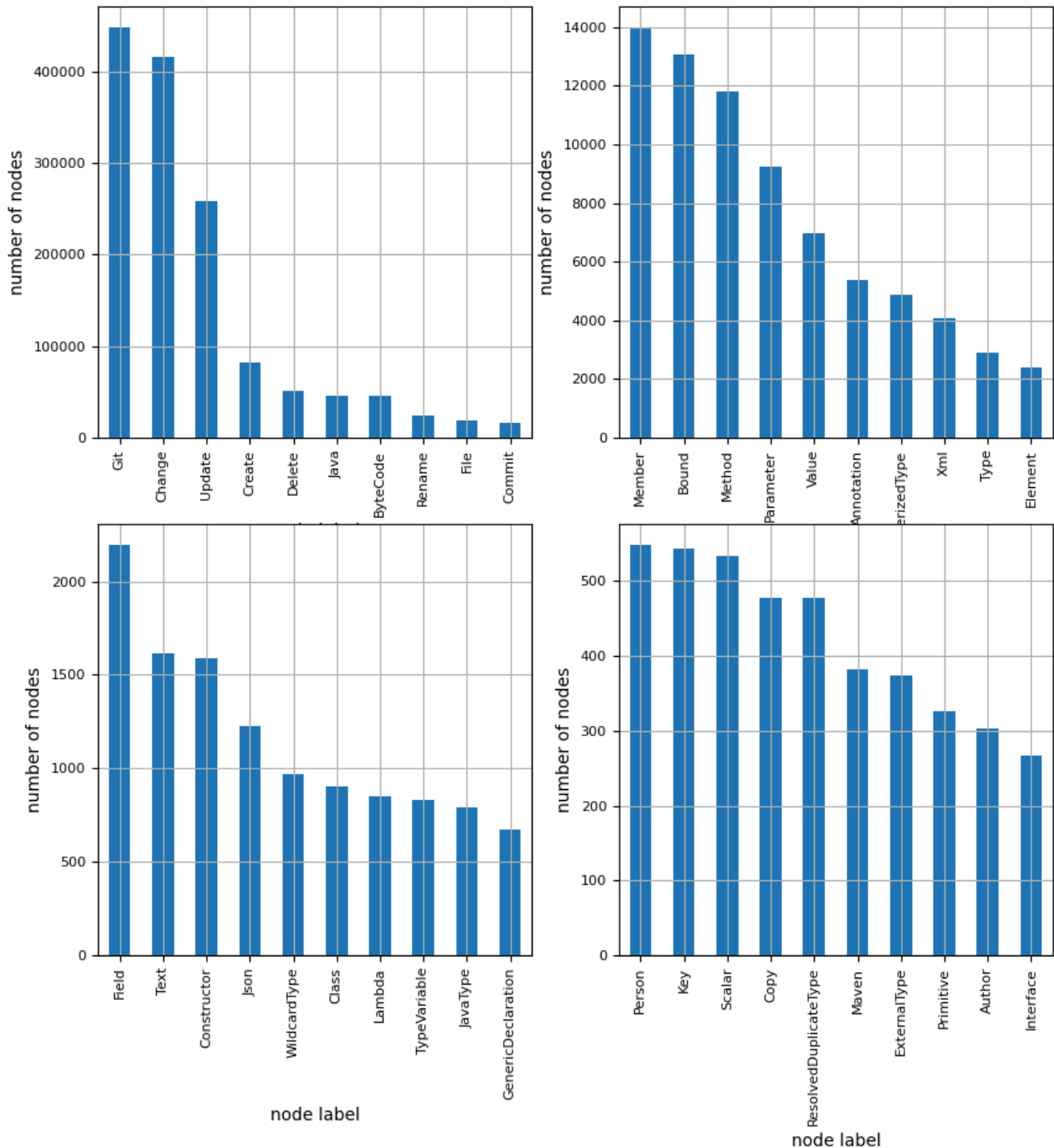
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	448647	89.707870
1	Change	416186	83.217228
2	Update	259019	51.791370
3	Create	82087	16.413461
4	Delete	50533	10.104175
5	Java	45315	9.060825
6	ByteCode	45114	9.020635
7	Rename	24069	4.812645
8	File	19205	3.840078
9	Commit	15712	3.141646
10	Member	14004	2.800128
11	Bound	13065	2.612373
12	Method	11807	2.360833
13	Parameter	9258	1.851156
14	Value	6969	1.393466
15	Annotation	5393	1.078341
16	ParameterizedType	4854	0.970567
17	Xml	4093	0.818404
18	Type	2888	0.577461
19	Element	2404	0.480685
20	Field	2197	0.439295
21	Text	1612	0.322323
22	Constructor	1586	0.317124
23	Json	1223	0.244541
24	WildcardType	969	0.193753
25	Class	900	0.179957
26	Lambda	851	0.170159
27	TypeVariable	831	0.166160
28	JavaType	788	0.157562
29	GenericDeclaration	672	0.134368
30	Person	549	0.109774
31	Key	544	0.108774
32	Scalar	534	0.106774
33	Copy	478	0.095577
34	ResolvedDuplicateType	478	0.095577
35	Maven	382	0.076382
36	ExternalType	373	0.074582
37	Primitive	326	0.065184
38	Author	303	0.060585
39	Interface	266	0.053187

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1553324

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	416186	26.793251
1	MODIFIES	416186	26.793251
2	UPDATES	259019	16.675143
3	CREATES	106634	6.864891
4	DELETES	74602	4.802733
5	COMMITTED	31424	2.023016
6	RENAMES	24069	1.549516
7	INVOKES	22257	1.432863
8	HAS_PARENT	19094	1.229235
9	OF_TYPE	18557	1.194664
10	DEPENDS_ON	17631	1.135050
11	HAS_FILE	16055	1.033590
12	HAS_COMMIT	15712	1.011508
13	DECLARES	14346	0.923568
14	OF_RAW_TYPE	13136	0.845670
15	HAS	9882	0.636184
16	HAS_NEW_NAME	9396	0.604896
17	RETURNS	8126	0.523136
18	READS	5952	0.383178
19	HAS_ACTUAL_TYPE_ARGUMENT	5758	0.370689
20	ANNOTATED_BY	5350	0.344423
21	OF_GENERIC_TYPE	4397	0.283070
22	CHANGED_TOGETHER_WITH	3982	0.256353
23	SIMILAR	3380	0.217598
24	CONTAINS	3194	0.205624
25	RESOLVES_TO	2597	0.167190
26	RETURNS_GENERIC	2590	0.166739
27	HAS_FIRST_CHILD	2404	0.154765
28	HAS_LAST_CHILD	2404	0.154765
29	HAS_ELEMENT	2382	0.153349

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

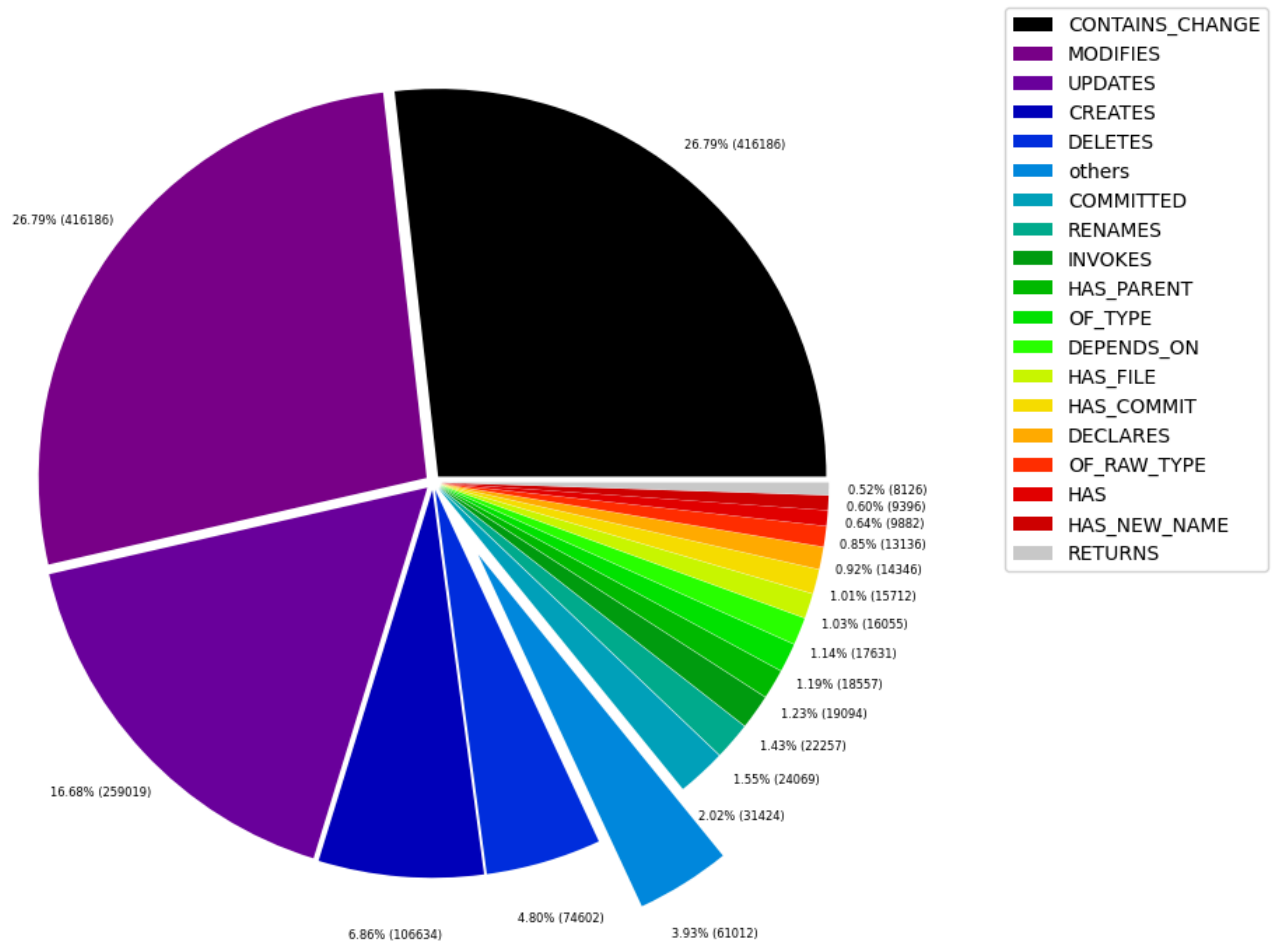


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

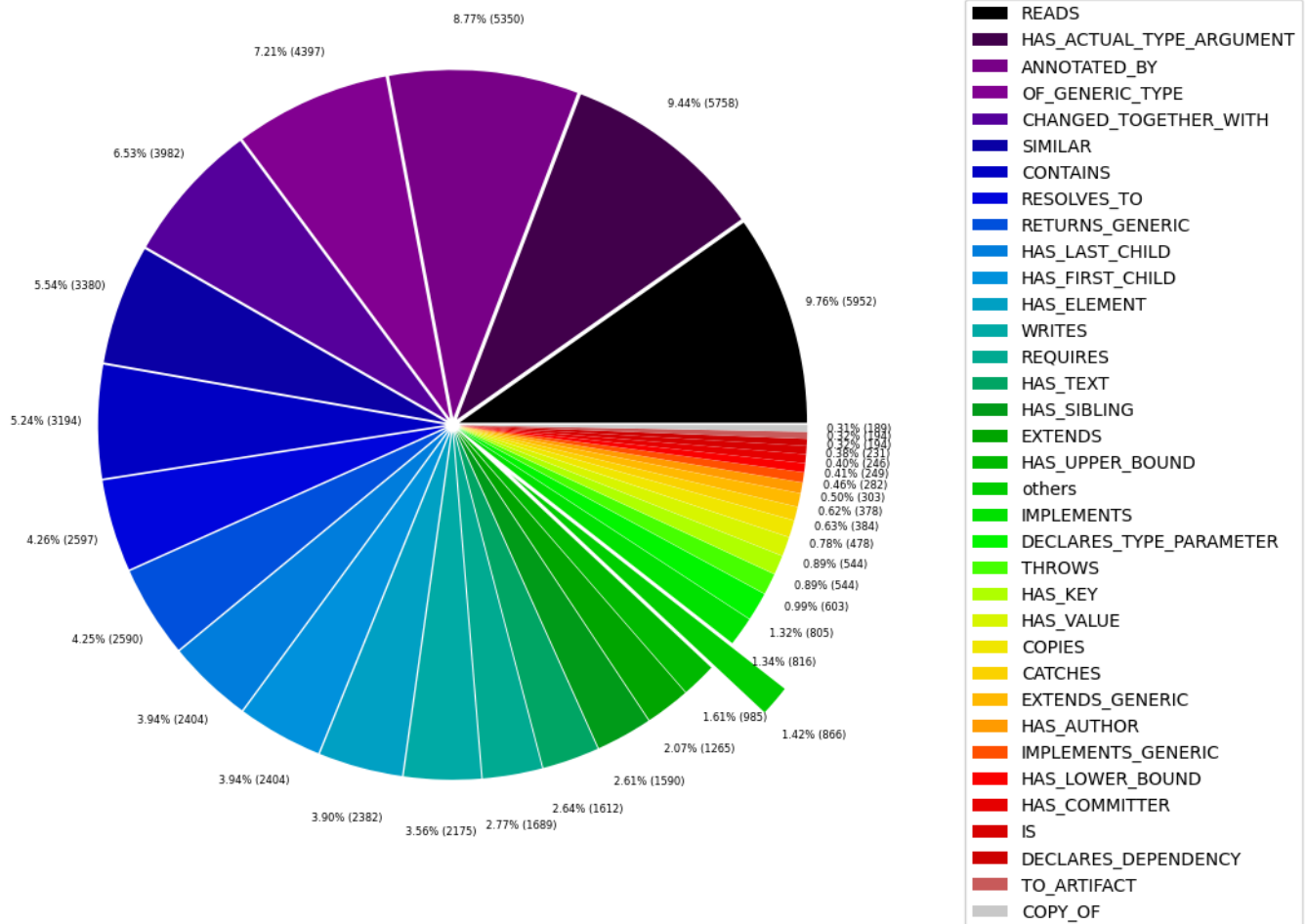
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000064
1	HAS_BRANCH	1	0.000064
2	HAS_HEAD	2	0.000129
3	THROWS_GENERIC	3	0.000193
4	EXCLUDES	4	0.000258
5	HAS_SCM	11	0.000708
6	DESCRIBES	11	0.000708
7	HAS_ROOT_ELEMENT	13	0.000837
8	HAS_GOAL	13	0.000837
9	HAS_EXECUTION	13	0.000837
10	INCLUDES_CONCEPT	19	0.001223
11	IS_ARTIFACT	20	0.001288
12	HAS_CONFIGURATION	20	0.001288
13	USES_PLUGIN	20	0.001288
14	OF_NAMESPACE	22	0.001416
15	HAS_ATTRIBUTE	22	0.001416
16	REQUIRES_TYPE_PARAMETER	26	0.001674
17	REQUIRES_CONCEPT	28	0.001803
18	DECLARES_NAMESPACE	44	0.002833
19	HAS_DEFAULT	53	0.003412
20	HAS_COMPONENT_TYPE	103	0.006631
21	CONTAINS_VALUE	131	0.008434
22	HAS_TAG	143	0.009206
23	ON_COMMIT	143	0.009206
24	COPY_OF	189	0.012167
25	TO_ARTIFACT	194	0.012489
26	DECLARES_DEPENDENCY	194	0.012489
27	IS	231	0.014871
28	HAS_COMMITTER	246	0.015837
29	HAS_LOWER_BOUND	249	0.016030

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOfNodes
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	259019	15712	
1	[Git, Update, Change]	MODIFIES	[File, Git]	259019	259019	
2	[Git, Update, Change]	UPDATES	[File, Git]	259019	259019	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	82087	15712	
4	[Git, Change, Create]	CREATES	[File, Git]	82087	82087	
5	[Git, Change, Create]	MODIFIES	[File, Git]	82087	82087	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	50533	15712	
7	[Git, Change, Delete]	MODIFIES	[File, Git]	50533	50533	
8	[Git, Change, Delete]	DELETES	[File, Git]	50533	50533	
9	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	24069	15712	
10	[Git, Change, Rename]	DELETES	[File, Git]	24069	24069	
11	[Git, Change, Rename]	RENAMES	[File, Git]	24069	24069	
12	[Git, Change, Rename]	MODIFIES	[File, Git]	24069	24069	
13	[Git, Change, Rename]	CREATES	[File, Git]	24069	24069	
14	[Git, Commit]	HAS_PARENT	[Git, Commit]	19083	15712	
15	[Repository, File, Git]	HAS_FILE	[File, Git]	16055	1	
16	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	15712	1	
17	[Author, Git, Person]	COMMITTED	[Git, Commit]	15712	303	
18	[Committer, Git, Person]	COMMITTED	[Git, Commit]	15712	246	
19	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	13251	8883	
20	[File, Git]	HAS_NEW_NAME	[File, Git]	9396	16055	
21	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	5351	8883	
22	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	5311	8883	
23	[Java, Value, ByteCode, Annotation]	OF_TYPE	[Type, File, Java, ByteCode, ExternalType, Ext...	5006	5351	
24	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	4114	9258	
25	[Java, ByteCode, Parameter]	ANNOTATED_BY	[Java, Value, ByteCode, Annotation]	3805	9258	
26	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	3349	16055	
27	[Java, ByteCode, ParameterizedType, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	2804	4854	
28	[Xml, Element]	HAS_ELEMENT	[Xml, Element]	2382	2404	
29	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	2260	6308	

Graph Density

total_number_of_nodes (vertices): 500120

total_number_of_relationships (edges): 1553324

-> total directed graph density: $6.210327108908058 \times 10^{-6}$

-> total directed graph density in percent: 0.0006210327108908057